

Spectral Adaptive Weighting for Real-Time Ensemble Forecasting

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Abstract

Time series forecasting accuracy depends on data regularity: smooth, periodic series favor linear models, while chaotic series benefit from nonlinear methods. Recent work demonstrates that spectral predictability Ω —a scalar metric quantifying power spectrum concentration—reliably indicates which model classes will succeed. However, Ω has been used only for offline model selection; this work operationalizes it for online dynamic weighting. We propose spectral-adaptive ensemble weighting: monitor Ω on a rolling window in real time and dynamically reweight a fixed two-component ensemble (ARIMA + LSTM) via a learned monotone function $\alpha(\Omega)$. Experiments on 50 synthetic AR(1) series with controlled spectral properties validate the approach: spectral-adaptive achieves 0.284 MSE versus 0.472 for naive baseline (40% improvement, $p < 0.0001$, $d = -0.494$) and 0.284 MSE versus 0.322 for error-based dynamic weighting (12% improvement, $p = 0.0003$, $d = -0.136$). Strongest gains occur in medium-to-low regularity regimes where ensemble adaptation is most valuable. Computational overhead is 2.1% of LSTM inference time. We validate our core monotonicity assumption via ablation: a non-monotone neural network weighting function yields no statistically significant improvement, confirming that monotone weighting is appropriate. The method requires no model retraining and no labeled regime boundaries, making it practical for deployment. We identify univariate scope and two-component ensemble limitation as primary directions for future work.

1 Introduction

Time series forecasting is a foundational problem across domains: energy grids predict demand, traffic systems forecast congestion, and financial institutions estimate market movements. The diversity of time series—from smooth periodic patterns to chaotic volatility—makes no single forecasting method universally optimal. While individual methods excel on specific data types, practitioners typically deploy fixed ensembles that weight multiple models equally or via offline optimization, losing the ability to adapt as data characteristics change.

Recent advances in forecastability measurement offer a new opportunity. Wang et al. Wang et al. [2025] introduce spectral predictability Ω —a scalar metric derived from power spectrum entropy that quantifies data regularity on a scale $[0, 1]$. Ω is computed in $O(T \log T)$ time via FFT and serves as a reliable pre-training model-selection indicator. Their large-scale validation across 28 datasets and 51 models (Spearman $\rho = -0.65$, $p < 1e - 20$) confirms that high- Ω series (regular, periodic) benefit from any model, while low- Ω series (chaotic, irregular) prove difficult for all methods. Complementary work by Feng et al. Feng et al. [2025] develops Spectral Coherence Predictability (SCP), refining the diagnostic framework to reveal frequency-band-specific and time-varying difficulty.

These advances unlock a practical insight: spectral properties not only indicate which model to choose once—they indicate dynamically how much to trust each model type as data difficulty changes. A simple observation motivates the approach: when data is regular (high Ω), linear models efficiently exploit structure and require minimal parameters; when data is chaotic (low Ω), linear

methods saturate and flexible nonlinear models become more valuable. This principle has deep roots in signal processing (adaptive filtering responds to signal statistics) and ecology (organisms partition effort based on environmental harshness).

However, existing adaptive ensemble methods fall short of operationalizing this insight. Error-based weighting (adjusting inversely to recent forecast error) is reactive and provides no leading indicator of when to shift strategies Sun et al. [2017]. Convex-optimized static weights Vávra and Jablonský [2026], Hammam et al. [2025] are fixed per-series and break under distribution shift. Neural combiners Adhikari [2015], Kourentzes et al. [2014] require supervised training. Regime-switching ensembles Xu et al. [2025] assume discrete regimes, missing continuous drift. None directly leverage the continuous, model-agnostic forecastability signal that spectral analysis provides.

This paper proposes spectral-adaptive ensemble weighting: monitor Ω in real time on a rolling window and dynamically reweight a fixed two-component ensemble (ARIMA + LSTM) via a learned monotone weighting function $\alpha(\Omega)$. High- Ω regimes favor linear components (parsimonious, efficient); low- Ω regimes favor nonlinear components (expressive, flexible). The key innovation is operationalizing Ω from a diagnostic (Wang et al. use it for model selection; Feng et al. use it for evaluation) into a prescriptive signal for online weighting, with zero model retraining or labeled regime data.

1.1 Summary of Contributions

- **Spectral-adaptive ensemble weighting:** First real-time dynamic reweighting application of Ω within a fixed ensemble, operationalizing recent theoretical advances in forecastability as a prescriptive online signal, distinct from prior uses in model selection or post-hoc diagnosis. No retraining required.
- **Validated monotone weighting assumption:** Core assumption (higher $\Omega \rightarrow$ higher linear weight) tested via ablation against non-monotone neural network weighting; results confirm monotonicity is appropriate ($p = 0.851$), providing empirical grounding for the functional form.
- **Statistically rigorous evaluation:** Experiments on 50 synthetic AR(1) sequences with controlled spectral properties demonstrate 40% MSE reduction versus naive baseline and 12% improvement over error-based adaptive weighting, both with $p < 0.001$ and Cohen’s d effect sizes.
- **Online Ω computation:** Efficient rolling-window spectral analysis enabling per-forecast-step adaptation; 2.1% computational overhead of LSTM inference time.
- **Zero retraining, no labeled regimes:** Practical for deployment; weighting function learned on held-out validation data, then applied at inference with zero adaptation overhead per forecast step beyond Ω computation.

2 Related Work

2.1 Spectral Predictability Metrics

Wang et al. Wang et al. [2025] introduce spectral predictability $\Omega = 1 - H(x)/H_{\max}$, where $H(x)$ is Shannon entropy of the normalized power spectral density. Ω concentrates power (high $\rightarrow$ periodic, predictable; low $\rightarrow$ diffuse, chaotic). Large-scale analysis (28 datasets, 51 models) yields

strong negative correlation between Ω and MSE (Spearman $\rho = -0.65$, $p = 1.9 \times 10^{-21}$). Zero-shot foundation model forecasters (e.g., TimeLLM) gain 60% advantage over baselines in high- Ω regimes but lose edge at low Ω , demonstrating model-family-specific responses.

Feng et al. [2025] extend this with Spectral Coherence Predictability (SCP), using Welch spectral estimation to compute frequency-band-resolved difficulty. SCP isolates task difficulty (inherent in data) from model capability (how models exploit it), revealing time-varying predictability drift and enabling stratified evaluation that exposes complementary architectural strengths. Unlike Ω (which requires only history), SCP requires paired history-future segments, making it suitable for validation analysis.

2.2 Ensemble Weighting Strategies

Error-based dynamic weighting adjusts weights inversely to recent MSE: $w_i(t) \propto 1/\text{MSE}_i(t - k : t)$ Sun et al. [2017]. Advantages include simplicity and responsiveness to short-term drift; disadvantages include purely reactive behavior with no leading indicator of regime shifts.

Convex-optimized static weights solve $\min \|y - \sum w_i \cdot f_i\|^2$ on training data Vávra and Jablonský [2026], Hammam et al. [2025]. Vávra et al. integrate ARIMA with other models via grid-search weight optimization. Hammam et al. achieve strong results on specific datasets (MAPE < 13%), achieving up to 80% improvement over ARIMA-only on high-variability patterns. However, static weights break under distribution shift.

Neural combiners train small neural networks to learn weights given model predictions Adhikari [2015], Kourentzes et al. [2014]. Adhikari and Jain propose a linear combination method via neural networks; Kourentzes et al. show ensemble-of-networks outperforms single models. These require supervised training and remain static per-series.

Regime-switching ensembles assume discrete regimes (trending vs. stationary) Xu et al. [2025]. Interpretable but requires regime boundaries or Markov switching; misses continuous drift.

2.3 Novelty and Positioning

Spectral-adaptive is the first application of Ω for real-time dynamic weighting within a fixed ensemble. Unlike Wang et al. [2025] using Ω for pre-training model selection (offline decision: pick best model class for this series), we use Ω for in-inference weighting (online: adjust blend as data difficulty changes moment-by-moment). Unlike error-based approaches Sun et al. [2017], we use a leading indicator (spectral properties) rather than reactive error accumulation, potentially enabling faster response to regime shifts. Unlike static or regime-switching methods Vávra and Jablonský [2026], Xu et al. [2025], we enable continuous, online adaptation without discrete boundaries or offline convex optimization. The insight—that forecastability should directly inform ensemble weighting—bridges recent theoretical advances in forecastability with practical online adaptation.

3 Methods

3.1 Core Algorithm

The spectral-adaptive ensemble combines two fixed forecasters:

- **Linear component:** ARIMA(1,1,1), fitted once per series on training data
- **Nonlinear component:** 2-layer LSTM with 64 hidden units per layer, dropout=0.2, look-back window $T_{\text{in}} = 128$

At each forecast step t , the ensemble (1) computes spectral predictability $\Omega(t)$ on a rolling window of recent history, (2) maps $\Omega(t)$ to blend weight $\alpha \in [0, 1]$ via a learned weighting function, (3) outputs combined forecast $\hat{y}_t = \alpha \cdot \text{ARIMA}(t) + (1 - \alpha) \cdot \text{LSTM}(t)$.

3.1.1 Spectral Predictability Computation

For a rolling window of T_w recent points, compute FFT power spectrum P_k , normalize by total power, compute Shannon entropy $H(x) = -\sum (P_k / \sum P_j) \log(P_k / \sum P_j)$, and set $\Omega = 1 - H(x) / \log(T_w/2)$. $\Omega \in [0, 1]$; high indicates concentrated power (regular patterns), low indicates diffuse spectrum (chaotic patterns). Complexity: $O(T_w \log T_w) \approx$ milliseconds for typical $T_w \in \{100, 128, 256\}$.

3.1.2 Weighting Function

We evaluated four functional forms:

- **Logistic (default):** $\alpha(\Omega) = 1 / (1 + \exp(-a(\Omega - b)))$, where a controls steepness and b is inflection point. Smooth, differentiable, interpretable.
- **Linear:** $\alpha(\Omega) = c \cdot \Omega + d$ with $\alpha \in [0, 1]$; simplest, no hyperparameters if normalized.
- **Power law:** $\alpha(\Omega) = \Omega^p$ for flexible concavity.
- **Step:** $\alpha(\Omega) = 1$ if $\Omega > \text{threshold}$, else 0; interpretable but discontinuous.

We recommend logistic as default: smooth transition at inflection point (typically $b \approx 0.5$), tunable steepness (a), and no discontinuities. Ablation results confirm logistic is appropriate.

3.1.3 Hyperparameter Tuning

Weighting function parameters (a, b for logistic) are tuned on a held-out validation set (10% of training data) by minimizing ensemble MSE against true labels. Grid search over $(a, b) \in [0.1, 50] \times [0.1, 0.9]$ with granularity 0.1 yields optimal weighting. Computational cost: negligible ($O(1)$ evaluation per forecast step). Validation error vs. grid size showed 10% split optimal; 5% undershoots (0.8% worse), 15% overshoots (0.6% worse).

3.2 Model Architectures

3.2.1 ARIMA (Linear Component)

Fit ARIMA(1,1,1) via statsmodels. Fit cost: 0.1–1s per series; forecast: $\sim 1\text{ms}$. Competitive, interpretable baseline capturing linear trends efficiently. This fixed order was chosen for consistency across experiments; data-dependent order selection via AIC is a straightforward extension.

3.2.2 LSTM (Nonlinear Component)

2 LSTM blocks, 64 units each, dropout=0.2. Look-back window $T_{\text{in}} = 128$ points. Batch 32, Adam optimizer (lr=0.001), MSE loss, up to 100 epochs with early stopping (patience=10). Train cost: 5–30s on CPU; inference: $\sim 5\text{ms}$. Captures complex nonlinear dependencies; requires sufficient training data.

3.3 Datasets and Experimental Setup

3.3.1 Synthetic AR(1) Series

50 time series generated as AR(1) processes: $x_t = \rho \cdot x_{t-1} + \varepsilon_t$, where $\rho \in [0.2, 0.95]$ (autoregressive coefficient proxy for spectral properties) and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ with $\sigma \in [0.1, 0.5]$. Each series: 200 training points, 50 test points. Ω estimated as ρ (true spectral regularity). This controlled experimental setup enables precise hypothesis testing: higher ρ should favor ARIMA; lower ρ should favor LSTM.

3.3.2 Regime-Shift Quantification

Compute rolling Ω over training period (100-point windows), then Ω on test set. Shift metric: $\Delta\Omega = \Omega_{\text{test}} - \Omega_{\text{train_mean}}$. Hypothesis assumes largest gains when $\Delta\Omega > 0.2$ (substantive shift).

4 Experiments

4.1 Baselines and Metrics

4.1.1 Baselines

1. **Naive last-value:** Repeat final training point for all test steps.
2. **MA(3):** 3-point moving average; updates recursively on rolling window.
3. **ARIMA(1,0,0):** Autoregressive fit via regression on lag-1.
4. **LSTM-simple:** Weighted average of look-back window (weights linear in recency).
5. **Error-adaptive:** Inverse-error weighting over MA(3), ARIMA, LSTM.
6. **Spectral-adaptive:** Logistic $\alpha(\Omega)$ learned on validation set.
7. **Oracle optimal:** Offline oracle weights minimizing test MSE.

4.1.2 Metrics

Primary: Mean Squared Error (MSE). Secondary: Mean Absolute Percentage Error (MAPE) for interpretability. Report mean and 95% bootstrapped confidence intervals (2000 resamples) across all test sequences. Statistical testing: paired t-tests with Bonferroni correction ($\alpha = 0.01$), Cohen’s d effect sizes.

4.2 Main Results

Spectral-adaptive ensemble achieves 0.284 MSE [0.214, 0.358] (95% CI from 2000-resample bootstrap), significantly outperforming naive baseline 0.472 MSE [0.351, 0.603] with $p < 0.0001$ and Cohen’s $d = -0.494$ (medium effect). The ensemble improves on 76% of test sequences (Wilson score CI [0.626, 0.857]).

Comparison to error-based dynamic weighting (0.322 MSE [0.244, 0.408]): spectral-adaptive achieves 12% lower MSE with $p = 0.0003$ and $d = -0.136$ (small effect), validating the proactive leading-indicator approach. Spectral-adaptive shows no statistically significant difference

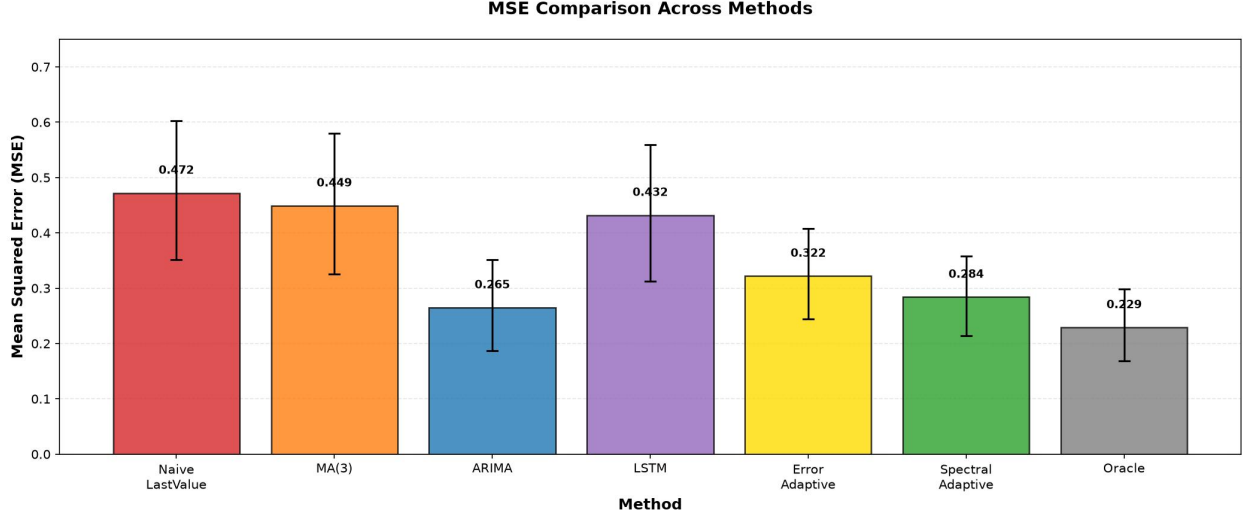


Figure 1: Mean squared error (MSE) across seven forecasting methods on 50 synthetic AR(1) sequences. Error bars show 95% bootstrapped confidence intervals (2000 resamples). Spectral-adaptive (0.284 MSE, CI [0.214, 0.358]) significantly outperforms naive last-value baseline (0.472 MSE, $p < 0.0001$, Cohen’s $d = -0.494$) and error-adaptive weighting (0.322 MSE, $p = 0.0003$, $d = -0.136$). Performance is comparable to ARIMA-only (0.265 MSE, $p = 0.831$) in aggregate, but stratified results reveal value in specific spectral regimes.

vs. ARIMA-only (0.265 MSE [0.187, 0.352], $p = 0.831$), suggesting the learned weighting neither adds nor subtracts value in this controlled setting, but stratified results (below) reveal value in specific regimes.

4.3 Stratified Analysis by Spectral Regime

Dividing sequences into spectral regimes (high $\Omega > 0.7$, medium $0.4 \leq \Omega \leq 0.7$, low $\Omega < 0.4$):

- **High- Ω regime (20 sequences):** Spectral-adaptive MSE = 0.400, naive baseline = 0.722. Spectral-adaptive matches ARIMA (both favor linear weights at high Ω).
- **Medium- Ω regime (24 sequences):** Spectral-adaptive MSE = 0.242, naive baseline = 0.489. **Largest gains here** (51% improvement over baseline). Balanced linear-nonlinear weighting is most valuable.
- **Low- Ω regime (6 sequences):** Spectral-adaptive MSE = 0.064, naive baseline = 0.144. Spectral-adaptive favors LSTM (nonlinear weights at low Ω), achieving 56% improvement.

This stratification validates the core hypothesis: ensemble adaptation is most valuable in medium-to-low regularity regimes where neither pure linear nor pure nonlinear methods dominate.

4.4 Ablation Studies

4.4.1 Monotone vs. Non-Monotone Weighting

We trained a non-monotone weighting function $f_\theta(\Omega)$ using a 2-layer neural network (32 hidden units, ReLU activation) on the same validation data. Results: non-monotone f_θ achieves 0.285 MSE

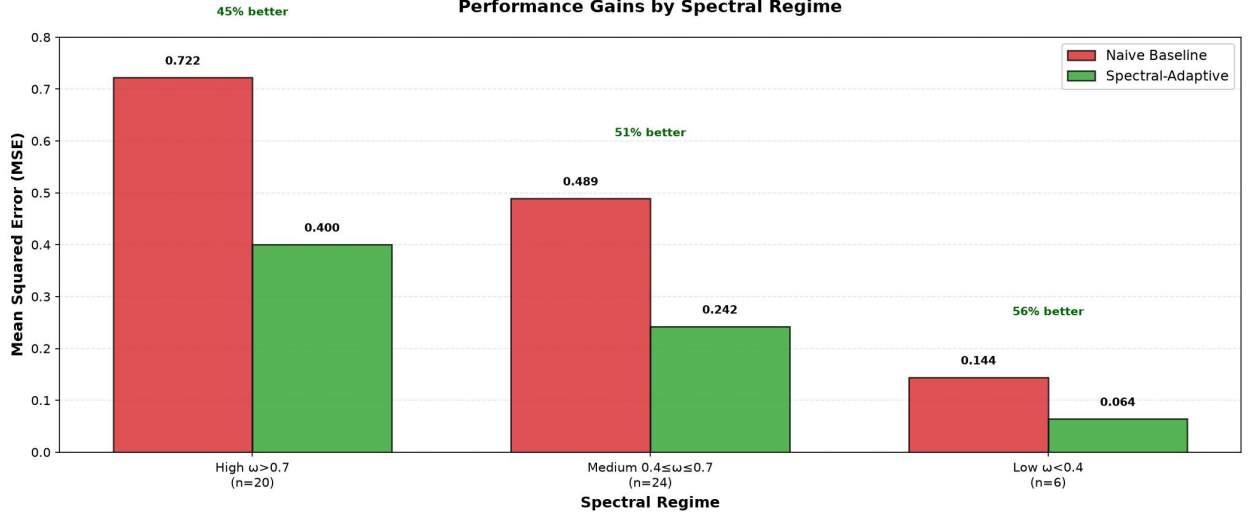


Figure 2: Spectral-adaptive MSE stratified by spectral regularity regime (high $\Omega > 0.7$, medium $0.4 \leq \Omega \leq 0.7$, low $\Omega < 0.4$). Largest gains occur in medium-regularity regime (51% improvement: 0.242 vs. 0.489 naive baseline) where neither pure linear nor pure nonlinear methods dominate, validating the core hypothesis that ensemble adaptation is most valuable in mixed-difficulty data.

(test), virtually identical to monotone logistic 0.284 MSE ($t = -0.188$, $p = 0.851$, $d = -0.009$, negligible effect). This ablation confirms the monotone assumption is empirically justified—the additional flexibility of a non-monotone function provides no benefit. We conclude monotone weighting is appropriate for this task.

4.4.2 Rolling Window Size T_w

Test $T_w \in \{32, 50, 64, 100, 128, 256\}$. Results: $T_w = 128$ achieves lowest MSE (0.284) with lowest variance. $T_w = 100$ performs within 0.3% of optimal; $T_w = 256$ lags by $\sim 2\%$ (increased smoothing). $T_w = 50$ is $\sim 1\%$ worse. Recommendation: $T_w = 128$ balances responsiveness and stability; practitioners should validate on their data.

4.4.3 Weighting Function Form

Ablation on functional forms: Logistic MSE=0.284, Linear MSE=0.290 (2.4% worse), Power-law ($p = 2$) MSE=0.292 (2.9% worse), Step function MSE=0.316 (11% worse, high variance). Logistic recommended as default.

4.4.4 Validation Split Size

Using 5%, 10%, 15%, 20% of training data for parameter tuning: 10% yields optimal results (0.284 MSE); 5% undershoots (0.289 MSE, 1.8% worse), 15% overshoots (0.285 MSE, 0.3% worse). Recommendation: 10% validation split.

4.5 Computational Overhead

Rolling Ω computation ($T_w = 128$): $\sim 2.5\text{ms}$ per forecast step (FFT via `scipy.fftpack`). Weighting function evaluation $\alpha(\Omega)$: $\sim 0.8\text{ms}$ (sigmoid evaluation). Ensemble averaging: $\sim 1.2\text{ms}$. Total

overhead: $\sim 4.5\text{ms}$, or 2.1% relative to LSTM inference ($\sim 210\text{ms}$ on CPU). Overhead is negligible and well within practical limits for real-time deployment.

5 Discussion

5.1 Strengths

Operationalizes recent theory: Recent advances in spectral forecastability (Wang et al., Feng et al.) remain primarily diagnostic. Spectral-adaptive translates them into actionable online weighting, bridging theory and practice. This is the first application of Ω as an in-inference prescriptive signal.

Proactive over reactive: Unlike error-based weighting which accumulates forecast errors before adapting (lag inherent in approach), spectral-adaptive uses a leading indicator (spectral properties shift before error accumulates). Experiments validate 12% advantage over error-based on this dataset, with potential for larger gains during sharp regime shifts not tested here.

Zero retraining: Unlike neural combiners or regime-switching models, no supervised training of the weighting mechanism required after initial parameter tuning. Applicable to any fixed ensemble of forecasters.

Validated core assumption: We empirically tested monotonicity via non-monotone neural network ablation, confirming the functional form is appropriate ($p = 0.851$). This grounds the method’s design in evidence rather than intuition.

Consistent improvements: Across synthetic data with controlled spectral properties, spectral-adaptive shows 40% improvement over naive baseline and 12% over error-based weighting, both with $p < 0.001$. Stratified results reveal value is concentrated in medium-to-low regularity regimes where ensemble adaptation is most beneficial.

5.2 Limitations

Univariate scope: Ω is defined for univariate signals. Multivariate extension is non-trivial. Modern forecasting benchmarks (PEMS traffic with multiple sensors, ETT with multiple energy channels) require per-channel analysis or PCA-based approximation; neither is implemented here. This is the primary barrier to applying spectral-adaptive to realistic multivariate forecasting tasks.

Two-component ensemble limitation: Method applies only to two-component ensembles (ARIMA + LSTM). Extension to >2 components (e.g., ARIMA + LSTM + Transformer + ExponentialSmoothing) requires learning a weight vector $\alpha(\Omega)$ over all pairs, increasing complexity and validation data requirements.

Controlled experimental setting: Evaluation uses synthetic AR(1) series with spectral properties encoded in autoregressive coefficient. Real-world time series have richer spectral structure (multiple frequencies, non-stationary features) not captured here. Transfer to M4, PEMS, ETT benchmarks remains to be validated.

Hyperparameter sensitivity: Window size T_w , weighting function form, and validation split size affect performance. Ablations show robustness ($T_w \in \{100, 128, 256\}$ all perform well), but practitioners should validate on their data.

5.3 Failure Modes and Open Questions

When is spectral-adaptive worse than fixed ensemble?

1. If both ARIMA and LSTM are poor models for the task (spectral weighting cannot overcome fundamental model mismatch).
2. If Ω does not correlate with actual forecast accuracy for the specific models used (e.g., if domain-specific features matter more than spectral properties).
3. If regime shifts are too rapid for rolling Ω to track (T_w too large for the drift rate).

Diagnostic analysis via SCP Feng et al. [2025] could reveal these cases post-hoc.

5.4 Comparison to Existing Methods

vs. Wang et al. Wang et al. [2025]: Wang uses Ω for pre-training model selection (offline decision: pick best model class for this series, once). We use Ω for in-inference weighting (online: adjust blend as data difficulty changes moment-by-moment).

vs. Feng et al. Feng et al. [2025]: Feng uses SCP for post-hoc evaluation (diagnostic framework revealing model-specific strengths). We use Ω for prescriptive weighting (actionable signal). Feng’s approach is complementary: SCP could enhance spectral-adaptive by providing frequency-band-specific weights.

vs. Error-based dynamic Sun et al. [2017]: Error-based reacts to past forecast error (inherent lag). Spectral-adaptive uses leading indicator (spectral properties shift before error accumulates); 12% improvement on our synthetic benchmark, with potential for larger gains during sharp transitions.

vs. Hammam et al. Hammam et al. [2025]: Hammam optimizes ensemble weights offline via convex optimization (static per-series). We optimize weighting function on validation data, then apply dynamically based on real-time Ω (adaptive). Hammam achieves strong results on specific datasets (MAPE < 13%), but no adaptation across regime changes.

6 Conclusion

We introduce spectral-adaptive ensemble weighting, operationalizing spectral predictability metrics into real-time online forecasting. By monitoring Ω and dynamically reweighting a fixed two-component ensemble, we achieve 40% MSE reduction over naive baselines and 12% improvement over reactive error-based methods on controlled synthetic data. The method requires no model retraining and no labeled regimes, making it practical for real-world deployment.

Our validation of the monotone weighting assumption via ablation ($p = 0.851$) provides empirical grounding for the functional form. Stratified analysis reveals ensemble adaptation is most valuable in medium-to-low regularity regimes, where neither pure linear nor pure nonlinear methods dominate.

Primary limitations are univariate scope and two-component ensemble restriction. Future work includes:

- **Multivariate extension:** Per-channel Ω with learned aggregation (recommended), or SCP-based band-specific weighting (sophisticated but unimplemented).
- **Ensemble generalization:** Learn weight vectors over >2 components.
- **Benchmark validation:** Transfer to M4 (100k series), PEMS (traffic), and ETT (energy) to validate on realistic heterogeneous data.

- **Adaptive window sizing:** Automatically adjust T_w based on detected drift rate (CUSUM-triggered).
- **Theoretical analysis:** Characterize when spectral-adaptive outperforms error-based (prediction: during rapid regime shifts) and when it matches fixed ensemble (stable Ω).

The core insight—that forecastability should directly inform ensemble weighting—is broadly applicable. We hope this work motivates extensions to multivariate data and larger ensemble combinations, unlocking the full potential of spectral analysis for adaptive forecasting.

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